



# **COMPUTER ORGANIZATION & ASSEMBLY LANGUAGE(CS-215T)**

**Lecture by:**  
**Mr. Shakir Karim**  
**Assistant Professor**  
**DoCS&IT, SSUET**

Batch 2022F



Week 11

# Computer Memory Types & Hierarchy, Cache Memory



# CONTENTS TO BE COVERED

1. Computer Memory Systems
2. Characteristics of Memory System
3. Why Computer Hierarchy?
4. Concept of Locality of Reference
5. Cache Memory and Element of Cache Design



# COMPUTER MEMORY SYSTEM

- Memory is
- A logically structured – linear array of locations
- Locations has addresses from 0 to maximum size a processor can address
- Classification is based on distance from the processor
- *number of machine cycles required for access*
- Closer to processor, Faster the memory



# CHARACTERISTICS OF MEMORY SYSTEM

|                                                                                                                                      |                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Location</b><br>Internal (e.g., processor registers, cache, main memory)<br>External (e.g., optical disks, magnetic disks, tapes) | <b>Performance</b><br>Access time<br>Cycle time<br>Transfer rate                |
| <b>Capacity</b><br>Number of words<br>Number of bytes                                                                                | <b>Physical Type</b><br>Semiconductor<br>Magnetic<br>Optical<br>Magneto-optical |
| <b>Unit of Transfer</b><br>Word<br>Block                                                                                             | <b>Physical Characteristics</b><br>Volatile/nonvolatile<br>Erasable/nonerasable |
| <b>Access Method</b><br>Sequential<br>Direct<br>Random<br>Associative                                                                | <b>Organization</b><br>Memory modules                                           |



Solution

# MEMORY HIERARCHY

## Hierarchy

To cater for the trade-off among three characteristics

How much?

i.e., Capacity?

How Fast?

i.e., Access Time?

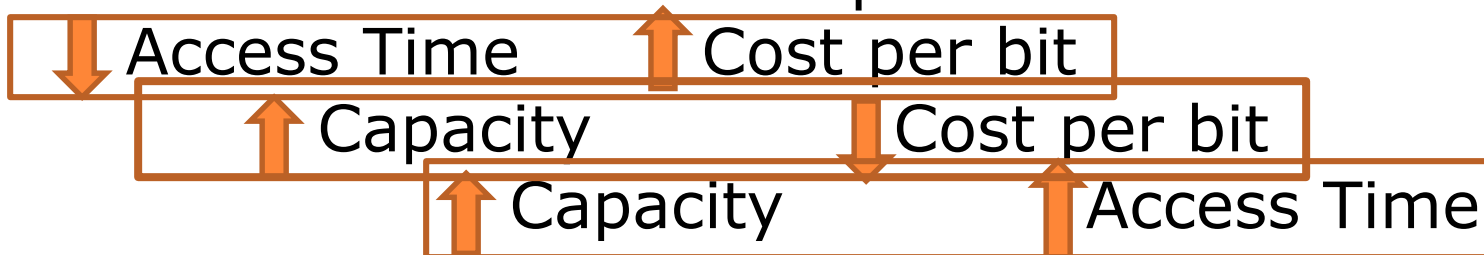
How Expensive?

i.e., Cost?

Dilemma exists!

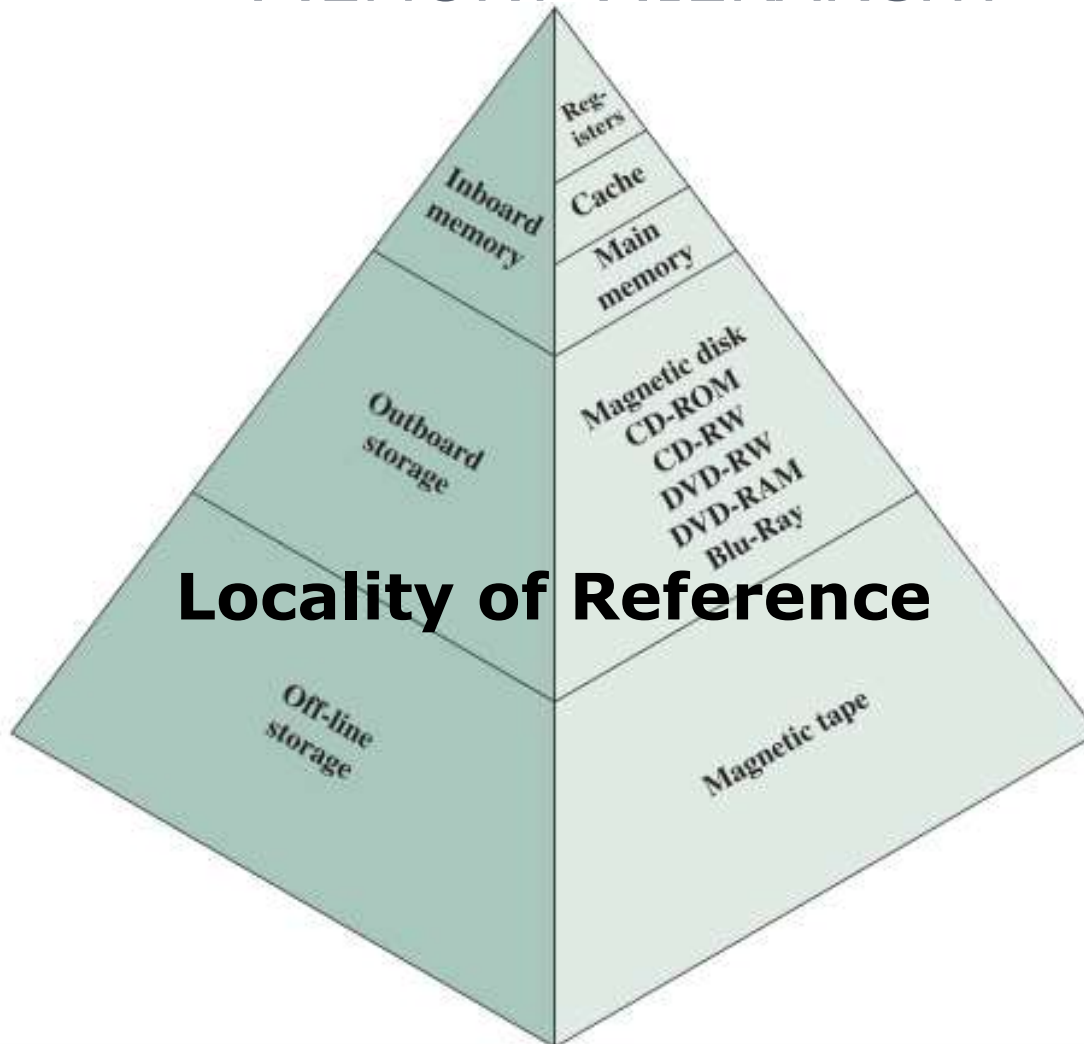


Possible trade-off relationships hold:





# MEMORY HIERARCHY



**Locality of Reference**

processor.



# TERMS RELATED TO MEMORY HIERARCHY

- *Hit*—requested data in a given level of memory (typically, only for upper levels of memory).
- *Miss*—requested data not in given level of memory.
- *Hit rate*—percentage of memory accesses found in a given level of memory.
- *Miss rate*—  $(1 - \text{Hit Rate})$
- *Hit time*—time required to access the requested information in a given level of memory
- *Miss penalty*—time required to process a miss,
  - that includes replacing a block in an upper level of memory,
  - **plus** the additional time to deliver the requested data to the processor.



## LOCALITY OF REFERENCE

- If memory location  $X$  is accessed at time  $t$ ,
- (assuming no branch instructions)
- there is a high probability that
- memory location  $X + 1$  will also be accessed in the near future.
- This clustering of memory references into groups is an example of

***Locality of Reference***



# LOCALITY OF REFERENCE – TYPES

- ***Temporal locality***

*Recently accessed items tend to be accessed again in the near future.*

- ***Spatial locality***

*Accesses tend to be clustered in the address space (for example, as in arrays or loops).*

- ***Sequential locality***

*Instructions tend to be accessed sequentially.*



## LOCALITY OF REFERENCE – BENEFIT

- *a small amount of very fast memory can be used to effectively accelerate the majority of memory accesses.*

### **REASON:**

- only a small amount of the entire memory space is being accessed at any given time
- values in that space are being accessed repeatedly.
- $\therefore$  values from a slower memory can be copied to a smaller but faster memory that resides higher in the hierarchy.



# LOCALITY OF REFERENCE – BENEFIT

## REASON:

- only a small amount of the entire memory space is being accessed at any given time
- values in that space are being accessed repeatedly.
- $\therefore$  values from a slower memory can be copied to a smaller but faster memory that resides higher in the hierarchy.
- **Results in a memory system that can store**
- A large amount of information in a large, low cost memory
- yet provide nearly the same access speeds that would result from using very fast but expensive memory.



# LOCALITY OF REFERENCE – EXAMPLE

**EXAMPLE 4.1** Suppose that the processor has access to two levels of memory. Level 1 contains 1000 words and has an access time of  $0.01 \mu s$ ; level 2 contains 100,000 words and has an access time of  $0.1 \mu s$ . Assume that if a word to be accessed is in level 1, then the processor accesses it directly. If it is in level 2, then the word is first transferred to level 1 and then accessed by the processor. For simplicity, we ignore the time required for the processor to determine whether the word is in level 1 or level 2.

Average time to access a word ?

In our example, suppose 95% of the memory accesses are found in level 1. Then the average time to access a word can be expressed as

$$(0.95)(0.01 \mu s) + (0.05)(0.01 \mu s + 0.1 \mu s) = 0.0095 + 0.0055 = 0.015 \mu s$$

The average access time is much closer to  $0.01 \mu s$  than to  $0.1 \mu s$ , as desired.



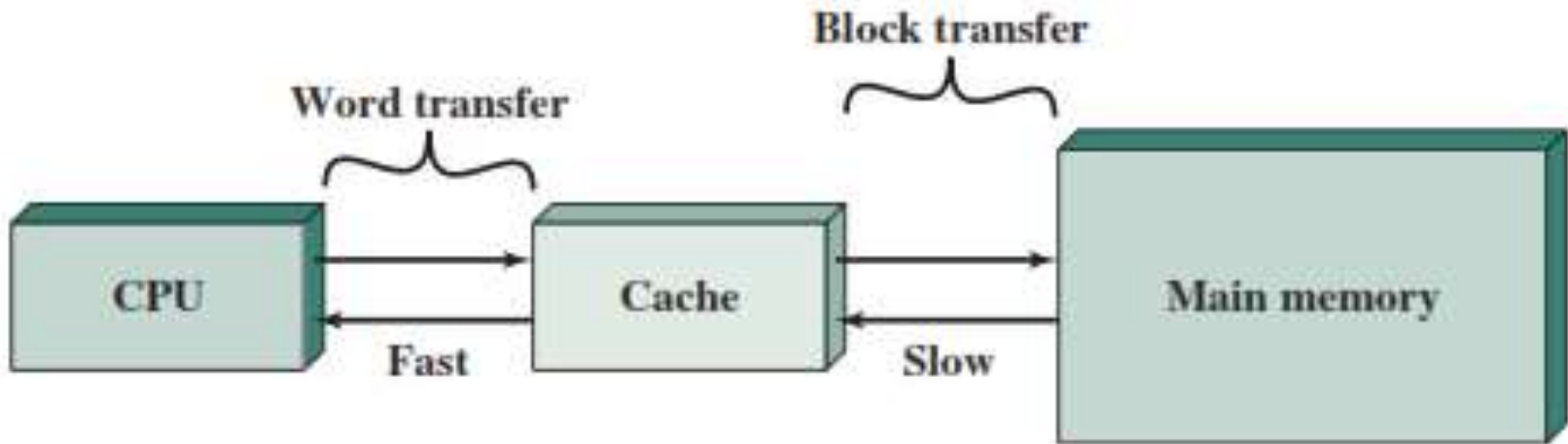
# CACHE MEMORY

- Cache Memory is
- a relatively small, fast memory
- Interposed between a larger, slower memory and the logic that accesses the larger memory.
- holds recently accessed data
- designed to speed up subsequent access to the same data.

*combines memory access time of expensive, high-speed memory with the large memory size of less expensive, lower-speed memory.*

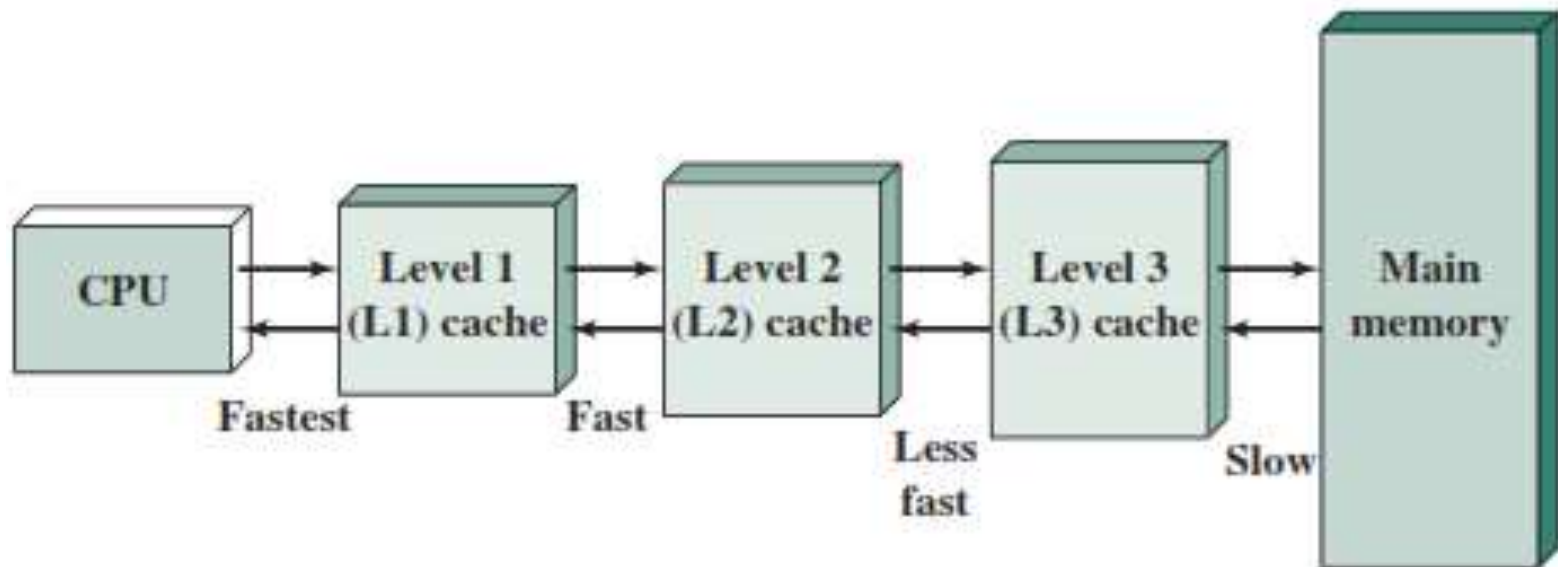


# CACHE MEMORY



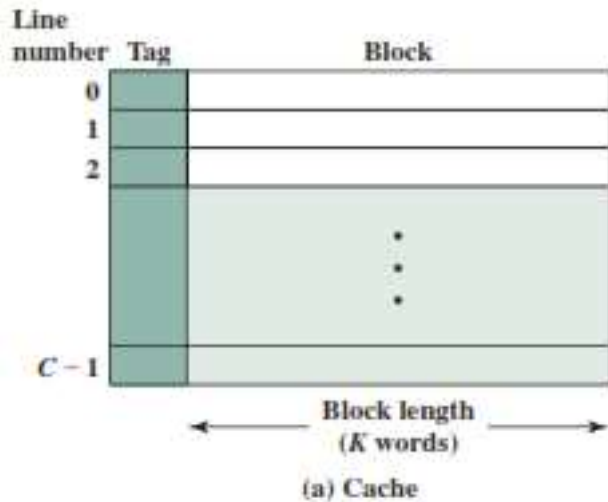


# CACHE MEMORY – ORGANIZATION

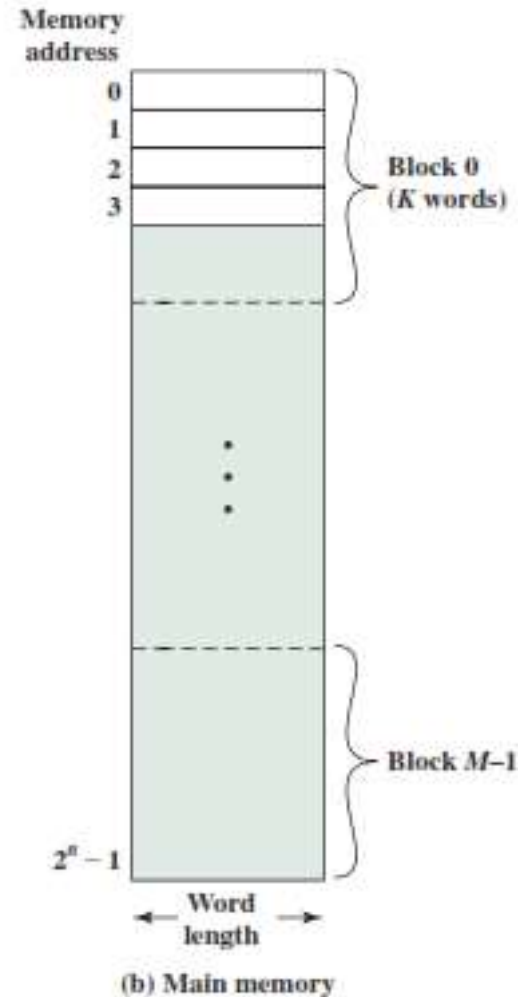




# CACHE MEMORY – STRUCTURE

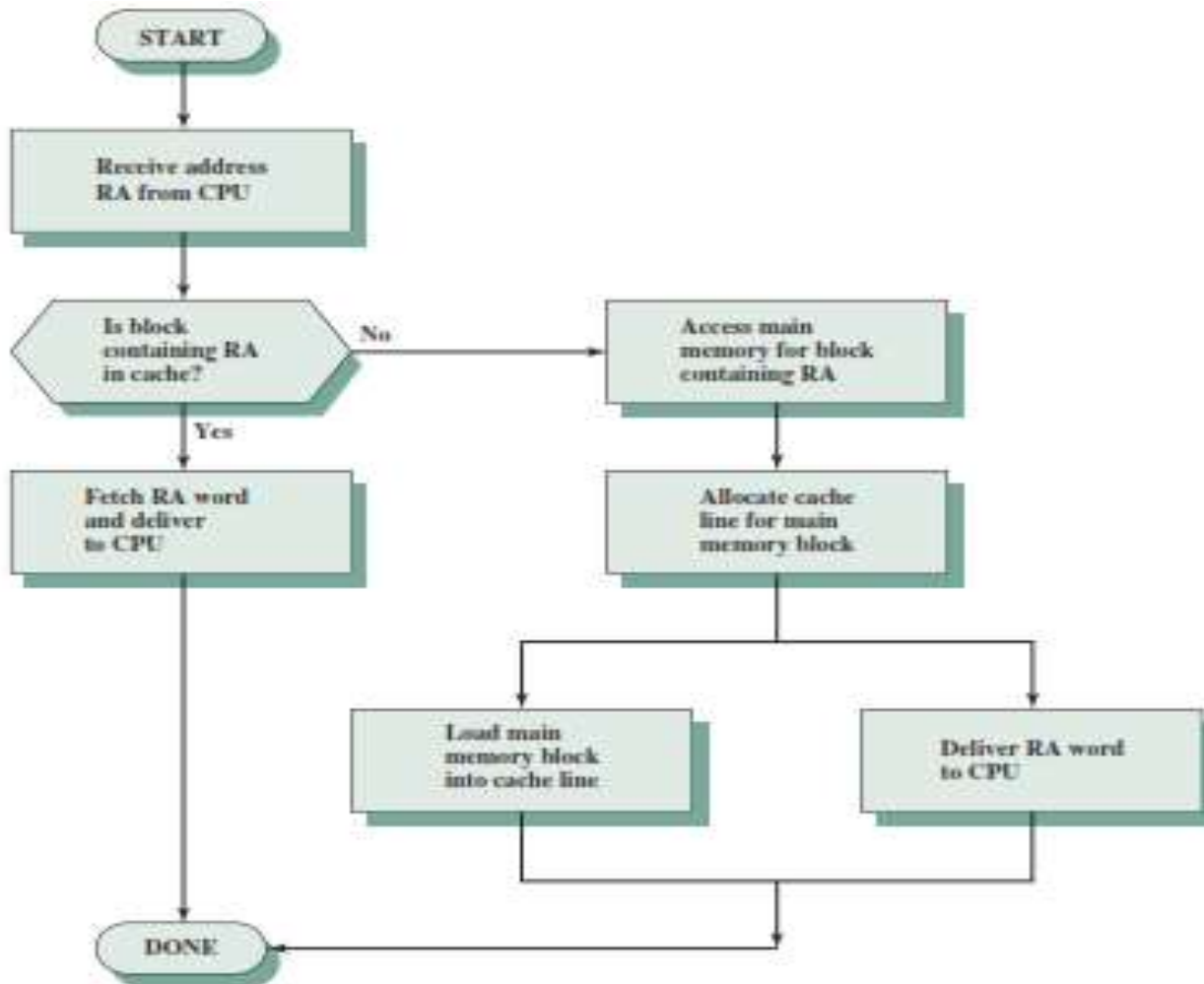


$$C \ll M$$



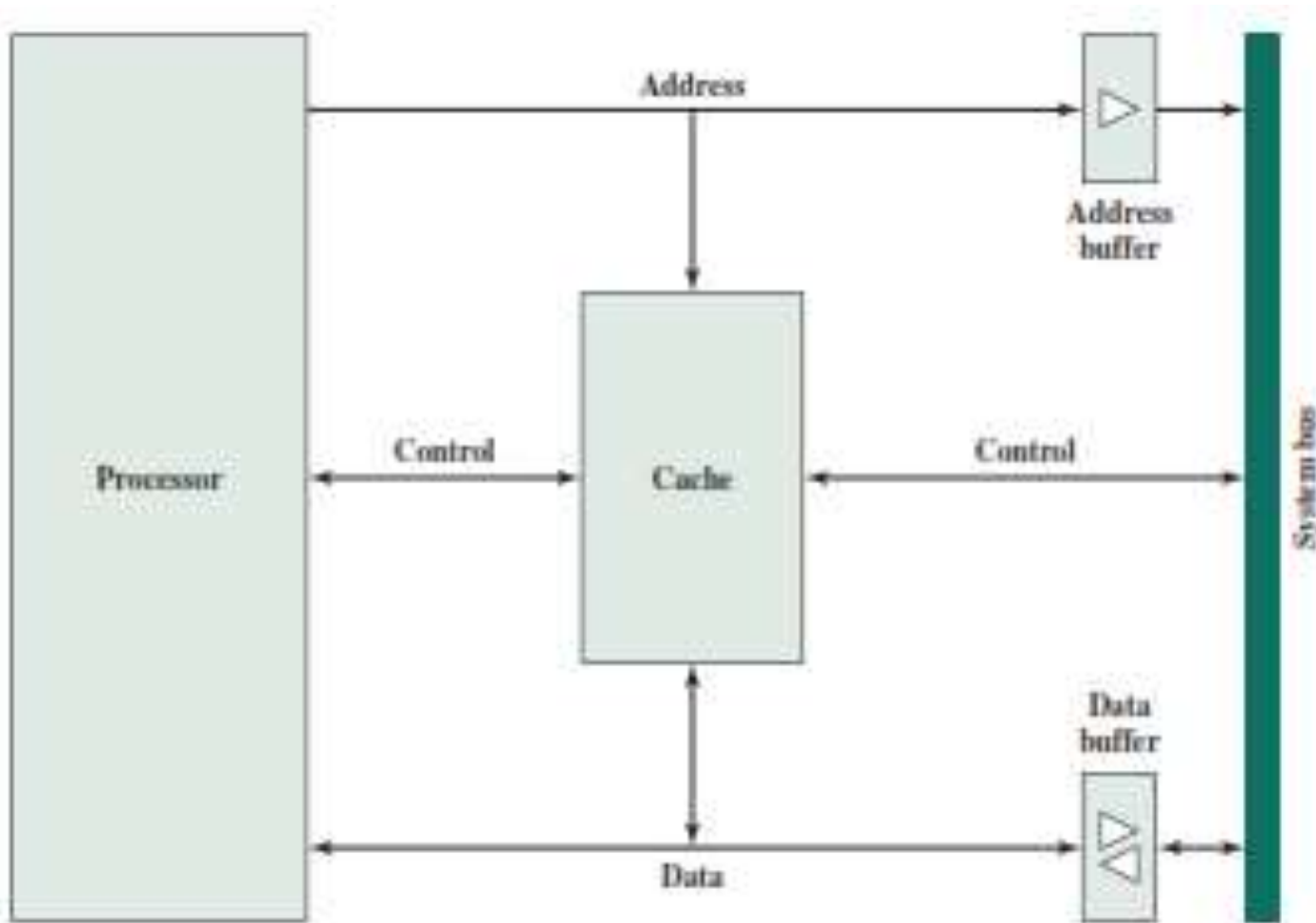


# CACHE MEMORY – READ OPERATION





# CACHE MEMORY – TYPICAL ORGANIZATION





# ELEMENTS OF CACHE DESIGN

## **Cache Addresses**

- Logical
- Physical

## **Cache Size**

## **Mapping Function**

- Direct
- Associative
- Set associative

## **Replacement Algorithm**

- Least recently used (LRU)
- First in first out (FIFO)
- Least frequently used (LFU)
- Random

## **Write Policy**

- Write through
- Write back

## **Line Size**

## **Number of Caches**

- Single or two level
- Unified or split



# ELEMENTS OF CACHE DESIGN

**EXAMPLE 4.2** For all three cases, the example includes the following elements:

- The cache can hold 64 kB.
- Data are transferred between main memory and the cache in blocks of 4 bytes each. This means that the cache is organized as  $16K = 2^{14}$  lines of 4 bytes each.
- The main memory consists of 16 MB, with each byte directly addressable by a 24-bit address ( $2^{24} = 16M$ ). Thus, for mapping purposes, we can consider main memory to consist of 4M blocks of 4 bytes each.